EXCELLENCE

Class 11 Physics · Session 2026–27

Units and Measurement

Clear formula notes — every word explained

This notebook is written for a student who wants every idea clear. First we explain in plain English. Then we explain every symbol. Then we show the formula on its own line. Then a small example.

How to use each page

1. Read “In plain English”

2. Read “What the words mean”

3. Copy the formula box

4. Try the green example

Chapter map

0. Symbol dictionary (read this first)
1. Number and unit
2. Dimensions (M, L, T)
3. Changing one unit into another
4. Checking if a formula is possible
5. Finding a formula using dimensions
6. Significant figures (which digits to keep)
7. Accuracy, precision and kinds of error
8. Absolute, relative and percentage error
9. How error grows when we use a formula
10. Metre scale, vernier, screw gauge
11. Revision lines to say aloud

0. Symbol dictionary — read this first

Many students get stuck only because a short letter is not explained. Keep this page open while you study the rest.

Symbol / short name	Full meaning in school language
Q	any physical quantity (mass, length, force…)
n	the number part (like 5 in 5 kg)
u	the unit part (like kg in 5 kg)
M	mass dimension (from kilogram)
L	length dimension (from metre)
T	time dimension (from second)
[Q]	dimensional formula of Q (powers of M, L, T)
Δx (delta x)	absolute error in x — how many units wrong
$\bar{x}$ (x bar)	mean = average of many readings
L.C.	least count = smallest step the tool can show
M.S.D.	main scale division = gap between two main marks
V.S.D.	vernier scale division = gap between two vernier marks
M.S.R.	main scale reading = last main mark before vernier zero
V.S.R.	vernier scale reading = vernier mark that matches a main mark
Pitch	how far a screw moves in one full turn
N	number of equal marks on the circular (round) scale

Remember: If a formula uses a letter you do not know, come back to this table first. Do not memorise a formula with unknown letters.

1. Number and unit

In plain English: whenever we measure something, we write two parts — how many, and of what. “5” alone is not enough. “5 kg” is complete.

What the words mean
- Physical quantity = the thing we measure (mass, length, time, force…)
- Number (n) = the digit part
- Unit (u) = the agreed size we compare with (kg, m, s…)
- SI = the world standard unit system used in school and exams

Physical quantity = number × unit

$$Q = n \times u$$

SI starts with 7 base quantities. Other units (newton, joule, watt) are built from these.

What we measure	SI unit name	We write
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Light intensity	candela	cd

Worked example — writing a quantity
• A bag has mass five kilograms.
• Number n = 5. Unit u = kg.
• So Q = 5 × kg.
Final answer: 5 kg

2. Dimensions — what sits inside a quantity

In plain English: dimension tells us which base things hide inside a quantity. Force is made from mass, length and time. We write that as powers of M, L and T.

What the words mean
- M = mass, L = length, T = time
- [force] means “dimensional formula of force”
- L T^{-1} means length to power 1, time to power −1 (like speed = distance/time)
- A negative power means “divided by” that quantity

Dimension = powers of M, L, T inside a quantity
We write it inside square brackets: [force], [speed]

Table to learn (most asked)

Read the middle column first. It tells you how to remember the left column.

Quantity	Think like this	Dimensional formula
Speed / velocity	distance ÷ time	$[v] = L\,T^{-1}$
Acceleration	speed ÷ time	$[a] = L\,T^{-2}$
Force	mass × acceleration	$[F] = M\,L\,T^{-2}$
Momentum	mass × speed	$[p] = M\,L\,T^{-1}$
Work / energy	force × distance	$[W] = M\,L^{2}\,T^{-2}$
Power	work ÷ time	$[P] = M\,L^{2}\,T^{-3}$
Pressure	force ÷ area	$[P] = M\,L^{-1}\,T^{-2}$
Density	mass ÷ volume	$[\rho] = M\,L^{-3}$
Frequency	1 ÷ time	$[f] = T^{-1}$
Angular speed ω	angle ÷ time (angle has no dimension)	$[\omega] = T^{-1}$
Torque	force × distance (turning)	$[\tau] = M\,L^{2}\,T^{-2}$
G	from $F = G\,m_1\,m_2 / r^2$	$[G] = M^{-1}\,L^{3}\,T^{-2}$

Same formula — different names

Work, energy and torque
all have $M\,L^{2}\,T^{-2}$
Same dimensions does NOT mean
they are the same quantity

No dimension?

Angle in radian: has unit, no M/L/T
Strain = change in length ÷ length
Inside sin, cos, e^{x}, log: treat as 1
“No dimension” ≠ always “no unit”

Worked example — find [force]
• Force = mass × acceleration.
• Mass → M. Acceleration → $L T^{-2}$.
• So $[F] = M \times L T^{-2} = M L T^{-2}$.
Final answer: $[F] = M L T^{-2}$

3. Changing one unit into another

In plain English: the real amount of stuff does not change when you change the unit label. If the unit becomes smaller, the number must become larger.

What the words mean
• n_1 = old number, u_1 = old unit
• n_2 = new number, u_2 = new unit
• $n_1 u_1 = n_2 u_2$ means “same physical amount”

$$n_1 u_1 = n_2 u_2$$
$$n_2 = n_1 \times (u_1 / u_2)$$

Remember: For area, square the length factor. For volume, cube it. 1 m = 100 cm, so 1 m^2 = 100^2 cm^2 = 10,000 cm^2.

Worked example — 5 m^2 into cm^2
• 1 m = 100 cm, so 1 m^2 = $(100)^2$ cm^2 = 10^4 cm^2.
• 5 m^2 = 5×10^4 cm^2.
Final answer: 5×10^4 cm^2

4. Checking if a formula is possible

In plain English: you can add only like with like. You cannot add metres to kilograms. So in a correct physics equation, every term that is added or subtracted must have the same dimensions. This rule is called homogeneity.

What the words mean
- Homogeneity = “same dimensions on both sides / every added term”
- A, B, C here are any physical terms in an equation
- [A] means dimensional formula of A

If A + B = C
then [A] = [B] = [C]

You can use this to

Check if a given formula can be correct

Find the dimension of an unknown letter

Find a missing power

For checking: treat +, − like =

Dimensions cannot

Find a pure number (2π, ½, 6π)

Tell work apart from torque

(they share the same dimensions)

Build a formula from angle alone

Remember: Inside sin( ), cos( ), exp( ), log( ), the thing inside must have no dimension. If the inside has length or time sitting alone, the formula is wrong.

5. Finding a formula using dimensions

In plain English: suppose time period T depends on length l and gravity g. We guess $T = k \times l^a \times g^b$. Then we match M, L, T powers on both sides and find a and b. The constant k (like 2π) is found from experiment, not from dimensions.

What the words mean
- ∝ means “is proportional to” (same as × a constant)
- a, b = unknown powers we must find
- k = a pure number (dimensions cannot find it)
- Equate = match powers of M, then L, then T

Write: $Q = k A^a B^b C^c$
Match powers of M on both sides
Match powers of L on both sides
Match powers of T on both sides
Solve for a, b, c

Pendulum example (step by step)

Assume: $T = k\, l^a\, g^b$

Result of matching: $a = 1/2$, $b = -1/2$

So $T = k\sqrt{(l/g)}$

Experiment / theory gives $k = 2\pi$

Worked example — pendulum powers
- Left: $[T] = T^1$. Right: l has L, g has $L\,T^{-2}$.
- $T^1 = L^a \times (L\,T^{-2})^b = L^{a+b}\,T^{-2b}$.
- Time: $-2b = 1 \rightarrow b = -1/2$. Length: $a + b = 0 \rightarrow a = 1/2$.

Final answer: $T \propto \sqrt{(l/g)}$

6. Significant figures — which digits to keep

In plain English: significant figures are the digits we trust in a measurement. We keep all sure digits, and we keep one last digit that may be a little unsure. More significant digits means a finer (more detailed) measurement.

What the words mean
- Significant digit = a digit that counts in the measurement
- Leading zeros (0.0025) only show place value — they do not count
- Trailing zeros after a decimal (2.500) do count — they were measured
- 10^3 in 2.5×10^3 is only a place label — not an extra digit

Rule in simple words	Example	How many digits?
Digits 1–9 always count	123	3
Zero between digits counts	1002	4
Zeros only for place do not count	0.0025	2
Zeros after decimal at end count	2.500	4
Power of ten is not a digit	2.5×10^3	2

After you calculate

Add or subtract:

keep the least number of decimal places

Multiply or divide:

keep the least number of significant figures

Rounding

Look at the next digit

If it is 0–4 → leave the keep-digit

If it is 5–9 → raise the keep-digit by 1

(school rule for exactly 5: raise by 1)

7. Accuracy, precision and kinds of error

In plain English: accuracy means your answer is close to the true value. Precision means your repeated readings are close to each other. You can hit the same wrong spot every time — that is precise, but not accurate.

What the words mean
- True value = the correct value (or the best accepted value)
- Error = difference between your reading and the true value
- Systematic = always biased the same way (wrong zero, wrong scale)
- Random = sometimes high, sometimes low (eye, shake)

Kind of error	What it looks like in the lab
Systematic error	Every reading is shifted the same way
Random error	Readings scatter up and down around the mean
Least count error	Tool cannot show smaller than its smallest mark
Gross error	A human mistake (wrong writing, wrong scale used)

8. Absolute, relative and percentage error

In plain English: absolute error says “how many units wrong”. Relative error says “what fraction is wrong”. Percentage error says “how many parts in 100 are wrong”.

What the words mean
- Δx (say “delta x”) = absolute error in x
- measured = the value you got
- true = the correct / accepted value
- x in Δx/x is usually the true value (or the measured value if true is not known)
- Absolute error has the same unit as x. Relative error has no unit.

Absolute error: Δx = |measured − true|
Relative error: Δx / x
Percentage error: (Δx / x) × 100

How to write the final answer
(value ± absolute error) unit
Example: (5.0 ± 0.1) cm

Many readings of one quantity

Step 1: find the mean (average)

Step 2: each error = |reading − mean|

Step 3: mean absolute error =

average of those errors

Write (mean ± mean absolute error)

Only the tool is known

If the question gives no better info,

take absolute error ≈ least count

Metre scale → about ± 1 mm

Vernier → ± its least count

Worked example — percentage error
• True length = 50.0 cm. Measured = 49.0 cm.
• Absolute error $\Delta x = |49.0 - 50.0| = 1.0$ cm.
• Percentage error = $(1.0 / 50.0) \times 100 = 2\%$.
Final answer: 2%

9. How error grows when we use a formula

In plain English: if A and B both have errors, and you make Y from them, Y also has error. The rule depends on what you do with A and B.

What the words mean
• Y = the final quantity you calculate
• ΔY = absolute error in Y
• ΔY/Y = relative error in Y
• |m| means take the power as a positive number (for maximum error)
• Exact numbers like ½, 2, π add no error of their own

Four separate rules (learn one at a time)

If $Y = A + B$ or $Y = A - B$
then $\Delta Y = \Delta A + \Delta B$
(add the absolute errors)

If $Y = A \times B$ or $Y = A \div B$
then $\Delta Y/Y = \Delta A/A + \Delta B/B$
(add the relative errors)

If $Y = A^m \times B^n$
then $\Delta Y/Y = |m|(\Delta A/A) + |n|(\Delta B/B)$
(power multiplies that relative error)

If Y = (exact number) × A
then ΔY/Y = ΔA/A
(the exact number adds no error)

Remember: Memory trick: + or − → start with absolute error. × , ÷ or power → start with relative error. Then multiply relative error by Y if the question asks for absolute error.

Worked example — error in kinetic energy
• m = (5.0 ± 0.5) kg, v = (20.0 ± 0.4) m/s. Formula K = (1/2) m v².
• K = 1000 J. Relative error = 0.5/5 + 2×(0.4/20) = 0.10 + 0.04 = 0.14.
• Absolute error ΔK = 0.14 × 1000 = 140 J.
Final answer: (1000 ± 140) J

10. Metre scale, vernier and screw gauge

In plain English: least count is the smallest length the tool can show. Zero error is the false reading when nothing is being measured. We always remove that false reading from the reading we took.

What the words mean
- Least count (L.C.) = smallest step the instrument can show
- Main scale = the fixed long scale
- Vernier scale = the short sliding scale
- M.S.R. = main scale reading (mark just before vernier zero)
- V.S.R. = vernier scale reading (which vernier mark lines up)
- Pitch = distance the screw moves in one full turn
- Zero error = reading when jaws are closed with nothing inside
- Zero correction = opposite of zero error

Tool	How to find least count	How to take reading
Metre scale	usually 1 mm	reading ± 1 mm
Vernier	L.C. = 1 M.S.D. − 1 V.S.D.	M.S.R. + (V.S.R. × L.C.)
School vernier (10 marks)	often L.C. = 0.1 mm	same reading rule
Screw gauge	L.C. = Pitch / N	linear reading + (round mark × L.C.)

Reading taken = M.S.R. + (V.S.R. × L.C.)
True reading = reading taken − zero error
Zero correction = − (zero error)

Positive zero error

Jaws closed

Vernier zero is to the RIGHT of main zero

Zero error is positive (+)

Size = (matching vernier mark) × L.C.

Negative zero error

Jaws closed

Vernier zero is to the LEFT of main zero

Zero error is negative (−)

Size = −(N − matching mark) × L.C.

N = total vernier marks (often 10)

Worked example — vernier least count
- 25 vernier marks match 24 main marks. One main gap = 1 mm.
- 1 V.S.D. = 24/25 mm = 0.96 mm.
- L.C. = 1 − 0.96 = 0.04 mm.

Final answer: 0.04 mm

Worked example — true reading with zero error
- Reading taken = 8.56 cm. Zero error = +0.02 cm.
- True reading = 8.56 − 0.02 = 8.54 cm.

Final answer: 8.54 cm

11. Revision lines — say these aloud

Close the book. If you can say these in your own words, you understand the chapter.

Topic	Say this in one breath
Quantity	Always write number × unit.
Dimension	Force is $M\,L\,T^{-2}$. Energy is $M\,L^2\,T^{-2}$.
Convert	$n_1 u_1 = n_2 u_2$. Same amount, new number.
Check formula	Every added term must have the same dimensions.
Derive	Assume powers, match M, L, T, solve.
Limit	Dimensions never give a pure number like 2π.
Sig figs	+/− → decimal places. ×/÷ → significant figures.
Accuracy	Close to true. Precision = readings close together.
Absolute error	Δx = |measured − true| (same unit as x).
Relative error	$\Delta x / x$ (no unit). ×100 gives percentage.
Error grow	+/− add Δ. ×/÷/power add relative errors.
Vernier L.C.	1 M.S.D. − 1 V.S.D.
Vernier reading	M.S.R. + V.S.R. × L.C.
True reading	reading taken − zero error.
Screw gauge	L.C. = Pitch / N.

Last tip

If a sentence uses a short name you forgot, open page 0 (symbol dictionary). If you cannot say a formula in one simple sentence, that page needs one more read.

From Excellence Units lectures 01–06. For revision and class notes — not a question paper.